When Search Is Not Enough: Diagnosing Failure Before Constructing Verified Reasoning Capabilities

Chimdi Walter Ndubuisi

University of Missouri · ARC Prize Research Summit, 23 October 2026 · poster abstract

Many reasoning systems respond to failure by generating more programs, refining candidates or adding test-time computation; these strategies are bounded by the representations and operations available to search. We study a complementary question: can a reasoner tell whether a failure reflects insufficient search, an excluded representation, a missing executable capability or insufficient evidence, and adapt only that component without compromising the evidence used to judge it?

We present CORA, a framework for failure-conditioned construction and verification of task-local executable extensions. A candidate is built from auditable failure traces, installed provisionally into the unchanged native reasoner, and checked by demonstration replay, adaptive leave-one-out rebuilt per fold, independent target evaluation and removal controls; abstention is a legitimate outcome.

A frozen synthetic experiment produced 27 of 30 complete end-to-end witnesses (12 of 30 for the earlier engine) within its declared grammar. On 60 real ARC-AGI-2 training-development tasks, four automatically constructed growth relations compiled and installed (none before), and one was accepted and used by the native engine. None survived adaptive leave-one-out (0 of 14 folds), so no new ARC solve is claimed. Forensic analysis traced the dominant failure to representation admission. On particular two-demonstration subsets, finer segmentation views become eligible and fill a fixed four-view search cap, which cuts the still-eligible view holding the correct general program. On 24 synthetic control tasks, the obvious repair, forcing a view to be searched first, gained no task and harmed five, so it was not run. A fold-local alternative that removes duplicate views was also rejected (on 41 synthetic tasks it gained one and harmed four). Frozen neural guides chose the cut view no better: in a pre-registered synthetic test the fixed order chose correctly in 35 of 35 cases, Qwen2.5-1.5B in 19 (1 when also shown the real failure trace, as with a shuffled one) and I-JEPA in 3.

We outline a domain-portable control protocol, whose general failure-type controller is not yet implemented, and a prospective biological-perturbation test (untested). Neither biological transfer nor real ARC invention has been demonstrated.

Keywords: ARC-AGI-2; program synthesis; failure diagnosis; capability construction; verification; leave-one-out.

Evidence tiers: synthetic (frozen, run once); real development (training demonstrations only, no test output read); forensic (subsets examined after the failure was seen). The certified training score is unchanged at 185/1,000; a 120-task prospective pilot is frozen and has not been run. Every number is indexed with its commit and hash in `records/CORA_SUMMIT_POSTER_EVIDENCE_20261012.md`.